01. Extremal holomorphic CFT partition functions: certified necessary tests

Pure Thought research portfolio

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

Imagine trying to identify a musical instrument from a list of the notes it can produce. Some lists are impossible because the notes do not fit the instrument's mathematical rules. Others pass those checks, but that still does not tell us how to build the instrument. Here the "instrument" is a quantum theory and the "notes" are its allowed states. We ask whether exceptionally sparse lists of states obey the consistency rules. Such sparse theories are candidates for describing a simplified universe containing gravity with no additional matter.

Key terms

- **Conformal field theory (CFT)** — A quantum theory whose laws respect transformations that preserve angles, including changes of scale; it describes systems with no preferred length scale.
- **AdS_3** — A three-dimensional spacetime with a particular negative curvature, used as a mathematically tractable setting for gravity.
- **Holomorphic** — Depending on one complex coordinate rather than both it and its complex conjugate; this is a special restriction on the theories studied here.
- **Central charge c** — A number characterizing aspects of a CFT's symmetry and number of degrees of freedom. Here c=24k, with k a positive integer.
- **Partition function** — A generating function that packages the theory's state energies or scaling weights and the number of states at each level.
- **Primary and descendant** — A primary is a basic type of excitation; descendants are related excitations generated from it by symmetry.
- **Extremal spectrum** — A spectrum with the largest prescribed initial gap before additional primary excitations, under this particular ansatz.
- **Modular invariance** — The requirement that different equivalent descriptions of a torus give the same partition function.

Why this matters

Quantum gravity is hard partly because we do not know which mathematically plausible theories are internally consistent. These tests can eliminate candidates without experiments or a complete construction. They also connect gravity to the arithmetic of modular functions: counting quantum states becomes a problem about exact number sequences.

What progress would mean

A useful first result is a trustworthy table of candidates and any exact reason for rejecting one. Passing the table's tests would identify a candidate worth further study; constructing the full theory would remain a much stronger achievement.

Core question

For c=24k, which extremal holomorphic partition-function candidates pass exact genus-one spectral tests, and can any candidate be excluded by an additional, explicitly stated consistency condition?

Scope and assumptions

Start with bosonic holomorphic theories, a unique vacuum, c=24k, and k=1,…,4. This is a deliberately restricted version of the original gravity question. Use h for holomorphic weight; do not identify it with the full scaling dimension Δ=h+$\bar{h}$. The extremal ansatz has no nonvacuum Virasoro primary with h≤k. Treat the nonchiral modular bootstrap as a separate extension with independent (h,$\bar{h}$) data.

Keep the motivating existence question, but make the first deliverable a necessary-condition test. A modular function with integer coefficients is not a constructed CFT or a demonstrated gravity dual.

Mathematical target

With q=exp(2πiτ), c>1, use

$$\chi_0 = q^{-c/24}\prod_{n=2}^{\infty}(1-q^n)^{-1}, \qquad \chi_h = q^{h-(c-1)/24}/\eta(\tau) \quad (h>0).$$

Construct the unique degree-k polynomial in J=j−744 whose expansion agrees with χ_0 through q^0. Write $Z_k = \chi_0 + \sum_{h\geq k+1} d_h\chi_h$ and extract primary multiplicities exactly through a declared cutoff H, initially H=100. Prove modular invariance from the polynomial representation, not from evaluations at a few τ. A negative or noninteger d_h excludes the stated candidate; positivity through H is only a finite test.

Required outputs and proof obligations

Export the polynomial coefficients, exact q-series, primary multiplicities, conventions, cutoff, and a checker that independently expands η and J. An exclusion must name the violated necessary condition. An additional genus-two or OPE test must state its own assumptions and how its constraints follow from CFT consistency.

Validation and rejection controls

Recover J=q^{-1}+196884q+… at k=1 and d_2=196883 after subtracting the vacuum descendant. Test the checker with a deliberately altered coefficient. Check agreement of two series-construction routes. Do not report all-order coefficient positivity without a tail theorem.

Milestones and research extension

Baseline: exact k=1,…,4 tables with explicitly finite coverage. Strong: one additional sewing/OPE condition with independently checked consequences. Research extension: a new exclusion theorem or a full CFT construction; neither is a required outcome or a promised deadline.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Witten, Three-Dimensional Gravity Revisited — Motivation for the holomorphic extremal ansatz, not a proof that every candidate exists.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

For the holomorphic sector with $c = 24k$, $q = e^{2\pi i\tau}$ and $J = j - 744$,

$$\chi_0 = q^{-k}\prod_{n=2}^{\infty}(1-q^n)^{-1}, \qquad \chi_h = \frac{q^{h-k+1/24}}{\eta(\tau)} \quad (h>0).$$

$$Z_k = P_k(J) = \chi_0 + \sum_{h\geq k+1} d_h\chi_h, \qquad P_k(J) - \chi_0 = O(q).$$

Determine P_k exactly and test $d_h \in \mathbb{Z}_{\geq 0}$ through a declared H. At $k = 1$, $d_2 = 196883$. Passing this finite necessary test does not construct the CFT or establish OPE consistency.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. “Blocking” means the original issue can invalidate the scientific target or claimed certificate; “major” means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — incompatible sectors and gaps (original lines 16–29, 49–100).** The text alternates between holomorphic and diagonal nonchiral partition functions, then uses c/12 for both. For the chosen holomorphic extremal problem, the first permitted primary is h=k+1. The rewrite fixes one sector and separates h from Δ.
2. **Blocking — incorrect character identity (49–57).** The η expression drops the $q^{(1/24)}$ correction; the vacuum also requires removal of L_{-1}. These errors change the spectrum being optimized. The purported large-c S-matrix at 84–86 cannot serve as a universal Virasoro transformation law and is removed.
3. **Blocking — feasibility is not construction (18–23, 104 onward).** Continuous LP/SDP feasibility does not impose integer multiplicities, and neither a truncated spectrum nor genus-one consistency proves OPE associativity. The replacement identifies exactly what a successful computation establishes.
4. **Major — certificate coverage.** A positive grid or finite series does not control the omitted spectrum. The finite cutoff is an output, while all-order claims require a theorem. This also makes a useful baseline possible without solving extremal-CFT existence.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For c=24k, which extremal holomorphic partition-function candidates pass exact genus-one spectral tests, and can any candidate be excluded by an additional, explicitly stated consistency condition?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: exact k=1,…,4 tables with explicitly finite coverage. Strong: one additional sewing/OPE condition with independently checked consequences. Research extension: a new exclusion theorem or a full CFT construction; neither is a required outcome or a promised deadline.

Sources supporting the corrections

- Witten, Three-Dimensional Gravity Revisited — Motivation for the holomorphic extremal ansatz, not a proof that every candidate exists.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in

this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.